

Shawon Ghosh

Software Engineer

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Summary

Building scalable systems at the intersection of AI and software engineering from real-time computer vision pipelines to full-stack web platforms, across research labs and industry.

Education

Program	Institution	CGPA	Year
B.Tech. (Computer Science & Engg.)	National Institute of Technology Rourkela	7.36/10	2025
Higher Secondary Certificate (Science)	Govt. Debendra College, Manikganj	5.00/5.00	2019
Secondary School Certificate (Science)	Manikganj Govt. High School, Manikganj	5.00/5.00	2017

Professional Experience

Regnum Resource Ltd. <i>Software Engineer</i> - Developing intelligent transportation solutions powered by AI and Computer Vision. - Designing high-performance desktop applications using Python and PyQt for real-time monitoring. - Developing AI-based vehicle detection, tracking and classification modules using Deep Learning. - Building REST APIs with FastAPI and integrating Oracle SQL databases for enterprise applications. - Working on ANPR, RFID integration, traffic analytics and real-time automation systems. - Designing reusable, modular software architecture for large-scale intelligent systems.	Dhaka, Bangladesh <i>Jun 2026 – Present</i>
Betopia Limited <i>Technology Specialist</i> - Built enterprise business applications using Microsoft Power Apps with focus on scalability and usability. - Developed Visitor Management, Gate Pass and Requisition Management Systems, automating approval workflows. - Implemented workflow automation via Power Automate, reducing manual intervention significantly. - Administered Microsoft 365 and Active Directory (AD DS), managing users, access control and services. - Delivered scalable Power Platform solutions aligned with business and organizational requirements.	Dhaka, Bangladesh <i>Oct 2025 – May 2026</i>
Itransition Group <i>Intern Developer</i> - Developed and deployed scalable web applications using MERN stack and Agile practices. - Engineered a user management module with secure role-based authentication and authorization. - Built a library management system with automated fake data generation for testing. - Collaborated on a real-time presentation platform with multi-user editing and synchronization. - Delivered a customizable inventory system with dynamic data fields and adaptive visualization.	Minsk, Belarus <i>Jul 2025 – Sep 2025</i>
BigBasket <i>Software Engineer Intern</i> - Collaborated with the professional services team to integrate and optimize enterprise solutions. - Developed a workflow for store incharges to collect COD payments from delivery partners. - Built an interactive UI using ReactFlow to visualize and create reusable process and task templates. - Integrated the UI with existing Java Vert.x backend APIs to streamline workflow configuration.	Bengaluru, India <i>Jan 2025 – Jul 2025</i>
National Institute of Technology Rourkela <i>Undergraduate Research Assistant</i> - Conducted research on Building Classification under Diverse Lighting Conditions. - Applied image augmentation techniques: blur, rotation, zooming and histogram equalization. - Developed a CNN integrated with self-attention Transformer for robust image classification. - Published the research at an IEEE International Conference.	Rourkela, India <i>Jul 2024 – Dec 2024</i>
Indian Institute of Technology Mandi <i>Research Intern</i> - Worked in the Visual Intelligence and Machine Learning (VIML) research group. - Developed DICOMET, a desktop app for DICOM visualization, metadata extraction and report generation. - Improved AI detection accuracy and optimized bounding-box placement for medical image analysis. - Generated PDF reports and exported updated DICOM files for further medical research.	Mandi, India <i>May 2024 – Jul 2024</i>

Projects

Itemplate – Your Inventory, Your Way

Aug 2025

- Built a full-stack inventory system with **CRUD operations**, real-time updates, and role-based access control.
- Implemented **JWT, Google, and GitHub OAuth** authentication with secure API endpoints and rate limiting.
- Integrated **media uploads** via Multer + Cloudinary and optimized performance using pagination and compression.
- Designed a responsive UI with **React, TailwindCSS, React Query**, and multi-language support via i18next.

React, Node.js, Express, PostgreSQL, Socket.IO, TailwindCSS

Carrenter – Rent. Drive. Repeat

Mar 2024

- Developed a full-stack car rental platform with bookings and seamless user experience.
- Built a **Spring Boot** backend, optimizing data retrieval by **40%**.
- Implemented **JWT authentication** for secure login and user session management.
- Designed an admin dashboard to manage fleet and users, boosting operational efficiency.
- Optimized PostgreSQL queries to reduce data latency by **50%**.

React.js, Spring Boot, PostgreSQL, REST API, JWT Auth

NIRCTC Railway Reservation System

Nov 2023

- Built a GUI-based ticket booking system handling **1,500+** reservations.
- Integrated **SQLite3** for efficient data storage and search optimization (**45%** faster).
- Enabled interactive seat selection and automated email ticket confirmation.
- Applied **OOP principles** for better code organization and maintainability.

Python, Tkinter, CustomTkinter, SQLite3

Publications

- **S. Ghosh, S. K. Panda, M. K. Bishwal, and P. K. Sa**, “Transformer-Based Model for Building Classification Under Diverse Lighting Conditions,” in *Proceedings of the IEEE International Conference on Innovative Trends in Information Technology*, 2025. 

Technical Skills

Languages: Python, Java, C++, C, JavaScript, TypeScript, SQL

Frameworks: PyQt5, FastAPI, Flask, React.js, Node.js, Express.js, Vert.x

ML: PyTorch, TensorFlow, OpenCV, YOLOv8, DeepLabV3, Scikit-Learn, NumPy, Pandas

Databases: Oracle SQL, PostgreSQL, MySQL, MongoDB, SQLite

Cloud & Data: Microsoft Fabric, Power BI, Power Apps, Power Automate, SharePoint

Tools: Git, Docker, Linux, Firebase, MLflow

Certifications

- Microsoft Certified: Fabric Analytics Engineer Associate (DP-600)
- Microsoft Certified: Fabric Data Engineer Associate (DP-700)
- Microsoft Certified: SQL AI Developer Associate (DP-800)
- Microsoft Certified: Power BI Data Analyst Associate (PL-300)

Achievements

- Awarded the **ICCR International Scholarship** by the Government of India for pursuing a Bachelor of Technology at the National Institute of Technology Rourkela.
- Received the **General Scholarship** in recognition of consistent academic excellence throughout higher secondary education.

Positions of Responsibility

- Coordinator, NITRUTSAV, Annual Cultural Festival, NIT Rourkela.
- Volunteer, National Service Scheme (NSS), NIT Rourkela.

Languages

English (Professional), Bengali (Native), Hindi (Professional)